

Question:

A 74-year-old man comes to the physician for evaluation of a skin lesion on his right arm. The lesion first appeared 3 months ago and has since been slowly enlarging. Physical examination shows a 1.5-centimeter, faintly erythematous, raised lesion with irregular borders on the dorsum of the right forearm. A biopsy specimen is obtained. If present, which of the following histopathological features would be most consistent with carcinoma in situ? Answer Choices: A. Pleomorphism of cells in the stratum corneum B. Irreversible nuclear changes in the stratum basale C. Increased nuclear to cytoplasmic ratio in the stratum spinosum D. Presence of epithelial cells in the dermis

Reasoning + Answer:

<think> To approach this reasoning problem, we need to analyze the context of the question, which is centered around identifying the histopathological feature most consistent with carcinoma in situ, a form of skin cancer. Among the options, the correct answer is "Irreversible nuclear changes in the stratum basale," which refers to the abnormal and permanent changes in the cells at the base of the epidermis, a hallmark of in situ carcinomas. This directly addresses the pathophysiological process of skin cancer at its earliest, localized stage. </think> <answer> Irreversible nuclear changes in the stratum basale </answer>

